MST and shortest-path exercises

A08-M01 | Cuts and ties

For S={A,B,C}, list all crossing edges in the fixed graph. Let A={AB,AC,DE}. Does the cut respect A? Which edges are safe? Must every MST contain BD?

A08-M02 | Complete an exchange

Let A={AB,DE}, T={AB,BC,BD,DE,EF}, e=AC, S={C}. Give the cycle, a removable f, the reason $f \notin A$ and the weight equation. Which inference fails if the cut does not respect A?

A08-M03 | Strict cycle property

Use a triangle with all three weights equal to 1 to refute the claim that any maximum-weight cycle edge occurs in no MST. Then prove the statement for a uniquely heaviest edge.

A08-M04 | Every Kruskal round

Run all ten rounds on the fixed graph in (weight,edge name) order. Give acceptance, rejection and DSU components each round. For BC,CD,CE,DF,BF exhibit an already selected path. Internal DSU parent trees are not required.

A08-M05 | Every Prim round

Start from A. After each extraction give key/parent/Q in priority order. Identify the cut and a light edge before extracting D. Why does scanning CD keep parent[D]=B? Why can E acquire a negative key after D is extracted?

A08-M06 | Negative and disconnected

Keep only AB−2,AC1,BC1,DE−1,DF4,EF2. Find an MSF, its weight and edge count. Explain how Prim handles ∞. Does adding 10 to every edge weight change the optimal forest sets?

A08-M07 | Optimality and uniqueness

List all MST edge sets of the fixed graph. Prove that AB,DE,EF are mandatory. Give a three-vertex graph with repeated weights but a unique MST.

A08-M08 | Implementation costs

State Prim bounds and operation assumptions for indexed binary heaps, lazy heaps, array/matrix implementations and Fibonacci heaps. Separate Kruskal sorting and DSU costs. Why can a lazy heap exceed n entries?

Use graphs G+, G−, H, and N and the vertex/edge orders fixed in the slides. Slides contain every complete edge list.

A08-S01

Classify δ for s,a,b,c,u,v in N. Explain why the negative cycle on u,v does not affect s.

A08-S02

Before extracting b in G+, d[a]=4,d[b]=1,d[c]=∞. Relax b→a then b→c. Give labels and parents, and prove the upper-bound invariant.

A08-S03

Recompute every Dijkstra round on G+, including d, π, and Q. Identify when the parents of c and t change.

A08-S04

For H=s→a:2,s→b:5,b→a:−4,a→t:2, show Dijkstra failing and identify the broken inequality in its settlement proof.

A08-S05

Change a→c in G+ to −2. Process topological order s,b,a,c,t,z and reconstruct a shortest path to t.

A08-S06

Use the fixed seven-arc order in the slides to compute Bellman-Ford passes 0 through 5 on G−, including d and π. Why can topological edge order converge within one pass?

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A08-S07

Run five passes on N in its listed edge order. List relaxable arcs and the full procedure for assigning $-\infty$. Which answer is wrong if only cycle vertices are marked?

A08-S08

Prove the k-pass guarantee of Bellman-Ford and refute the claim that in-place pass k computes exactly the optimum over walks with at most k arcs. Compare time, space, and stale-entry handling for decrease-key and lazy binary heaps.

The shared graph G has these edges.

1→2=3, 1→3=8, 1→4=7, 2→3=-2, 2→4=5, 3→4=2, 4→1=2, 4→2=1.

A08-P01 States and bases

Distinguish $L^{(t)}$ from $D^{(k)}$. Why are their base matrices I and W? How must initialization handle a negative self-loop?

A08-P02 Min-plus and stopping

Prove $L^{(a)} \otimes L^{(b)} = L^{(a+b)}$ and associativity. Compute $L^{(2)}[1,4]$ and $L^{(3)}[1,4]$ on G. Why can budget n−1 miss a negative cycle?

A08-P03 Floyd stages k=2 and k=3

Starting from $D^{(1)}$ of G, compute $D^{(2)}$ and $D^{(3)}$. Identify all strictly improved cells and their predecessors. Explain by allowed sets why k=2 cannot yet reduce 1→4 to 3.

A08-P04 Loop order and predecessors

Trace why one increasing i,j,k pass misses 1→2 in the graph with only 1→4=1, 4→3=1, 3→2=1. Use $\Pi^{(4)}$ of G to reconstruct 3→1 and 1→4. Why copy $\Pi[k,j]$ rather than k?

A08-P05 Negative-cycle influence

A separate graph has only 1→2=1, 2→3=−2, 3→2=1, 3→4=2. List every $-\infty$ pair and find $\delta(1,1)$, $\delta(4,4)$, $\delta(4,1)$. State the reachability and negative-diagonal criterion.

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A08-P06 Johnson potentials and nonnegativity

Add the super-source to G. Compute h and all eight reweighted edges, then prove nonnegativity by the triangle inequality. Why can Bellman–Ford from 1 miss a disconnected negative cycle while the super-source detects it?

A08-P07 Order, restoration, and a false shortcut

Prove the telescoping formula. Restore original distances from $\delta(1,3)=3$ and $\delta(3,1)=2$. Give a counterexample to adding a constant C to each edge. Why can reweighting never remove a negative cycle?

A08-P08 Choose by density

For $m=\Theta(n)$ and $m=\Theta(n^2)$, compare time and output space for Floyd, Johnson with binary heaps, and Johnson with Fibonacci heaps. How does storing every Floyd stage change space?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapters 23–25. MIT 6.046J, Fall 2004, Lectures 13–16; Fall 2005, Lectures 16–19. Original handwritten algorithms notes (演算法_wjungle). These notes provide selected proof ideas and context; complete worked traces and diagrams are original educational synthesis.

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https://github.com/xinhi6716/computer-science-self-study